TRIBE 1 · THE BRAIN

The ML / AI Engineer

Computer Science & Modern AI

Core Strength:
Semantic Competence & Representation

- Vision-Language Models (VLMs)
- Action Chunking (ACT / Diffusion)
- Latent World Models (JEPAs)

The Critical Blindspot:

The Digital Sandbox Illusion—assuming simulation guarantees physical safety and crashes are harmless.

TRIBE 2 · THE NERVOUS SYSTEM

The Embedded / ECE Engineer

Silicon & Real-Time Systems

Core Strength:
Silicon Privilege & Multi-Rate IPC

- Microsecond Clocks & P_{99} Metrology
- Zero-Copy DMA & Shared SRAM
- Hardware Peripheral Bus Firewalls

The Critical Blindspot:

The Static Automation Illusion—treating real-time loops as rigid state machines unable to adapt.

TRIBE 3 · THE BODY & CONTROL

The Robotics Engineer

Dynamics, Invariants & Control

Core Strength:
Physical Laws & Safety Envelopes

- Multi-Body Inertia $\mathbf{M}(\mathbf{q})$ & Momentum
- Control Barrier Functions ($h(x) \geq 0$)
- Thermal Limits (I^2t) & Jerk Bounds

The Critical Blindspot:

The Closed-World Illusion—distrusting learned models as black boxes; fragile when domains drift.

Semantic Proposals (p_t) → **Multi-Rate IPC & Watchdogs** → **1 kHz CBF Safe Sets ($h(x) \geq 0$)**

THE PHYSICAL AI SYSTEMS SYNTHESIS

Bridging the Brain, Nervous System, and Body across the Proposal–Permission Boundary

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1. Unverified Proposals

High-capacity VLMs & Diffusion ACT emit candidate action chunks (p_t) on Linux MPU.

2. Real-Time Transport

Lock-free SRAM ring buffers and hardware watchdogs bound tail latency (P_{99}).

3. Physical Invariants

1 kHz MCU filters proposals onto safe set $\mathcal{U}_{\text{safe}}$ via CBFs before gate drive.

Universal Definition of Success: Open-World Semantic Competence AND Strict Physical Invariant Survival